ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE F

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Power transmission and gear ratios, open- and closed-loop systems, transducers, OR gates and reed switches, compressed-air supply and pneumatic circuits, a control circuit to study, and fault-finding, lock-out and sequential function charts.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all symbols and circuits neatly, using a ruler.
5. Use only a black or blue pen. Write neatly and legibly.

QUESTION 1: MECHANICAL AND TECHNICAL SYSTEMS

1.1 Name FOUR ways in which mechanical power can be transmitted from a motor to a machine. (4)

1.2 A driver gear with 20 teeth turns at 900 rpm and drives a gear with 60 teeth. Calculate the gear ratio and the speed of the driven gear. (5)

1.3 Explain the difference between an open-loop and a closed-loop technical system, and give ONE example of each. (6)

1.4 Explain what a transducer is, and give TWO examples. (5)

[20]

QUESTION 2: LOGIC AND SWITCHING

2.1 Draw the symbol and the truth table of an OR gate with inputs S1 and S2 and output E1. (5)

2.2 Explain how a magnetic reed switch works, and give ONE application. (4)

2.3 Explain the difference between a normally open (NO) and a normally closed (NC) contact. (4)

2.4 Name the FIVE steps of the problem-solving (design) process. (5)

2.5 Write the acronyms CNC and CAD in full. (2)

[20]

QUESTION 3: PNEUMATICS

3.1 Name FOUR components of a compressed-air supply system. (4)

3.2 Draw a pneumatic circuit in which a single-acting cylinder is controlled by a 3/2-way push-button valve with a spring return. Include the air supply. (6)

3.3 What is a two-pressure valve (AND valve) used for? Give ONE application. (4)

3.4 Draw the symbol of a 5/2-way valve that is pilot-operated on both sides. (4)

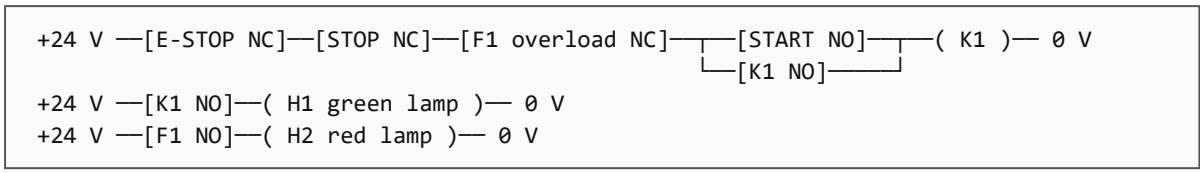
3.5 Explain why a double-acting cylinder produces a smaller force when it retracts than when it extends. (2)

[20]

QUESTION 4: ELECTRICAL CONTROL

4.1 Name FOUR devices that protect a circuit against over-current. (4)

4.2 Study the control circuit below and answer the questions that follow.



4.2.1 What is the purpose of the K1 contact connected in parallel with the START button? (2)

4.2.2 What happens when the emergency stop is pressed while the motor is running? (2)

4.2.3 What do lamps H1 and H2 show? (2)

4.2.4 Explain the operation of the circuit from the moment START is pressed until STOP is pressed. (6)

4.3 What is meant by a safety extra-low voltage (SELV), and how is it obtained from the mains supply? (4)

[20]

QUESTION 5: FAULT-FINDING AND MAINTENANCE

5.1 List THREE causes of air leaks in a pneumatic system. (3)

5.2 What is the purpose of a fault-finding flowchart? (2)

5.3 A pneumatic cylinder does not extend when its valve is operated. List, in a logical order, FOUR things you would check. (4)

5.4 Explain the lock-out procedure that must be followed before maintenance is done on a machine. (5)

5.5 Write SFC in full, and draw a simple SFC for a cylinder A that extends (A+) when START is pressed and retracts (A−) when it reaches its end position. (6)

[20]

TOTAL: 100

END OF QUESTION PAPER